

6.3 Friction Preference Pairing (FPP)

Friction Preference Pairing (FPP) adapts preference-based alignment to the epistemic setting by treating friction not as an explicit reward, but as a latent supervision signal that induces relative preferences over candidate responses (Pustejovsky and Krishnaswamy, 2025). In this paper, we refine this formulation using the productive/unproductive decomposition introduced in Section 5, which allows preference learning to distinguish epistemically beneficial hesitation from pathological or harmful friction.

FPP is motivated by a core limitation of reward-based shaping: many epistemic judgments are inherently comparative rather than absolute. Whether a response exhibits appropriate clarification, productive hesitation, or excessive deferral is often easier to assess relative to an alternative than to score on a fixed cardinal scale. Preference-based learning is therefore particularly well-suited to epistemic alignment, where the objective is not to assign a precise utility to each response, but to consistently prefer interventions that improve the long-horizon epistemic quality of an interaction.

This view therefore casts epistemic regulation as a *preference learning problem*. Instead of learning a scalar reward for friction, the model is trained to prefer interventions whose friction profile reflects epistemically productive behavior over those that reflect unproductive or pathological hesitation. This places FPP squarely in the tradition of preference-based alignment and direct preference optimization, while introducing a novel preference structure derived from the productive–unproductive friction decomposition.

Productive versus Unproductive Friction. As introduced in Section 5, we decompose friction as follows:

$$(19) \quad F(h, y) = F^+(h, y) - F^-(h, y)$$

where F^+ rewards epistemically productive interventions and F^- penalizes pathological hesitation.

This benefit–penalty decomposition mirrors standard formulations in preference-based and cost-sensitive learning, where desirable and undesirable attributes are separated to induce structured preferences (Ng and Russell, 2000; Christiano et al., 2017). Here it functions to separate friction that is epistemically beneficial (F^+ , e.g. clarification and information gain) from friction that reflects failure modes (F^- , e.g. contradiction, unsafe content,

miscalibration, or intent forcing). The minus sign indicates that unproductive friction is treated as a penalty.

Pair Construction. The central design question in FPP is how to induce a preference relation that reflects *epistemic quality* rather than surface helpfulness or verbosity. In many dialogue settings, whether an intervention is appropriate cannot be judged in isolation: a clarifying question may be preferable to a direct answer in one context, but worse in another. We therefore construct training data not from absolute scores, but from *pairwise comparisons* that explicitly contrast productive and unproductive uses of friction within the same epistemic context.

Our goal is to isolate preferences that reflect a normative distinction: interventions that increase short-term interaction cost in order to reduce long-term epistemic risk should be preferred to interventions that merely hedge, delay, or obscure without epistemic benefit. The following construction operationalizes this distinction by pairing responses according to their productive and unproductive friction profiles.

For a fixed dialogue context h , let y^+ and y^- denote the preferred and dispreferred responses, respectively, in a friction-based preference pair, constructed by sampling candidate responses $\{y_i\}_{i=1}^N \sim \pi_0(\cdot \mid h)$ and selecting two responses that satisfy the productive–unproductive friction ordering in Eq. (20).

$$(20) \quad \begin{aligned} \text{a. } & F^+(h, y^+) > F^+(h, y^-) \\ \text{b. } & F^-(h, y^+) < F^-(h, y^-) \end{aligned}$$

This construction is analogous to preference pair generation in inverse reinforcement learning and RLHF, where learning is driven by relative judgments rather than absolute scores (Christiano et al., 2017; Stiennon et al., 2020; Rafailov et al., 2023). Unlike standard preference learning, however, the preference relation here is not based on overall human utility, but on a structured epistemic decomposition of friction.

Specifically, y^+ is preferred to y^- when it exhibits strictly higher productive friction and strictly lower unproductive friction in the same context. This isolates *good epistemic hesitation* from mere verbosity, indecision, or defensive hedging.

Loss. We optimize a Direct Preference Optimization (DPO)-style logistic loss (Rafailov et al., 2023):

$$(21) \mathcal{L}_{\text{FPP}}(\theta) =$$

$$-\mathbb{E}_{(h, y^+, y^-)} \left[\log \sigma \left(\log \frac{\pi_\theta(y^+ | h)}{\pi_\theta(y^- | h)} \right) \right]$$

This is a pairwise preference objective: it increases $\pi_\theta(y^+ | h)$ relative to $\pi_\theta(y^- | h)$ using a logistic link, without learning an explicit scalar reward model. In FPO terms, the model is trained to prefer utterances whose friction profile is epistemically productive. In this way, the policy is directly optimized to satisfy epistemic preferences, avoiding the need to fit an explicit reward model and preventing the instability often associated with reward model miscalibration.

Discussion. Friction Preference Pairing represents a fundamentally different learning signal from reward shaping. Rather than inducing epistemic behavior indirectly through scalar returns, FPP trains the policy to satisfy a structured preference ordering over interventions.

The main advantage of FPP is that it avoids explicit reward modeling. This makes it robust to reward hacking and miscalibration, and well-suited to settings in which epistemic judgments are qualitative, comparative, or context-dependent. Because preferences are defined over pairs, FPP naturally handles intransitive and non-additive notions of epistemic quality that cannot be captured by a single scalar friction score. Critically, because FPP and GRFR rely on ordinal ranking rather than cardinal maximization, they mitigate the risk of exploiting holes in the underlying classifiers. The policy is driven to find 'better' interventions relative to a baseline, rather than exploiting the unbounded upper limits of a scalar reward model.

Moreover, FPP directly targets the intervention policy: the model is trained to prefer clarification, deferral, or refusal when these are epistemically productive, and to avoid pathological hesitation even when it superficially appears cautious.

The main limitation of FPP is that it requires either human annotations or reliable automatic heuristics to construct preference pairs. Its performance therefore depends on the quality of the productive-unproductive friction decomposition and on the consistency of the induced preference relation.

For these reasons, FPP is best suited to settings in which epistemic behavior can be meaningfully ranked by experts or curated heuristics, and serves as a complementary alternative to reward-based shaping in FAR.

6.4 Group-Relative Frictive Ranking (GRFR)

The first two FPO methods, FAR and FPP, operate primarily at the level of individual actions or single-turn preferences. While sufficient for local epistemic regulation, many failures of dialogue agents are inherently *trajectory-level*: excessive hesitation, delayed clarification, cascading contradictions, or gradual drift into unsafe behavior emerge only over multiple turns.

Group-Relative Frictive Ranking (GRFR) therefore casts epistemic alignment as an adaptation of group-relative policy optimization, extends trajectory ranking methods to epistemic intervention learning. GRFR learns to prefer entire dialogue strategies whose long-horizon epistemic conduct is superior relative to alternative strategies in the same context. This places GRFR in the tradition of rank-based and group-relative policy optimization, while introducing a novel epistemic ranking criterion derived from the friction functional. Here we provide a formal trajectory-level objective and a risk-sensitive decomposition of the scoring function.

Trajectory Groups. A central premise of GRFR is that many epistemic failures in dialogue are not local to a single turn, but emerge only over extended interaction: delayed clarification, cascading contradictions, gradual misalignment of beliefs, or escalation into unsafe regimes. Capturing such phenomena requires evaluating policies at the level of full dialogue trajectories rather than isolated responses.

Following standard formulations of dialogue management and sequential decision-making as partially observable Markov decision processes (Williams and Young, 2007; Young et al., 2013), we represent an interaction as a trajectory of intervention-utterance pairs over a finite horizon T :

$$(22) \tau_i = (a_{i,0}, y_{i,0}, \dots, a_{i,T}, y_{i,T}),$$

$$\text{where } i = 1, \dots, K$$

Here each trajectory τ_i corresponds to one possible dialogue strategy executed by the current policy in the same initial context h . Sampling multiple

trajectories per context induces a *cohort* of alternative epistemic strategies whose long-horizon consequences can be compared.

This grouped-trajectory formulation is standard in actor-critic and population-based reinforcement learning, where policies are evaluated relative to alternative rollouts to reduce variance and induce comparative learning (Sutton et al., 2000; Schulman et al., 2017). In the dialogue setting, it enables GRFR to reason explicitly about how sequences of clarifications, answers, refusals, and repairs interact over time.

Trajectories may include clarification turns, self-corrections, and epistemic repair actions, allowing GRFR to capture not only task success, but the coherence and stability of belief states across multi-turn interaction.

Trajectory Score. To rank full dialogue strategies, GRFR assigns each trajectory a scalar score that reflects both task performance and long-horizon epistemic conduct. Rather than evaluating individual actions in isolation, we score trajectories by accumulating per-step utility, intervention cost, and epistemic risk over time, following standard formulations of risk-sensitive and cost-sensitive control (Bertsekas, 1995; Tamar et al., 2015).

Formally, we define a trajectory-level score:

$$(23) \quad S(\tau) = \sum_{t=0}^T \left(R_{\text{task}}(h_t, a_t, y_t) - \lambda C_{\text{fric}}(a_t) - \eta \mathcal{R}_{\text{risk}}(h_t, a_t, y_t) \right) + \mu \sum_{t=0}^T F(h_t, y_t)$$

In this score, notice that the two epistemic terms in $S(\tau)$ serve distinct and complementary roles. The penalty $-\eta \mathcal{R}_{\text{risk}}(h_t, a_t, y_t)$ represents the agent’s *latent epistemic objective*: the unobservable ground-truth risk that the policy is ultimately intended to minimize. In contrast, the shaping term $\mu F(h_t, y_t)$ represents an *auxiliary, observable control signal* used to guide ranking and optimization when $\mathcal{R}_{\text{risk}}$ is partially delayed, sparse, or noisy. Such dual-objective formulations, combining a primary cost with an auxiliary shaping signal, are standard in reward shaping and multi-objective reinforcement learning (Ng et al., 1999; Wulfmeier et al., 2015).

Although F is constructed as a surrogate for $\mathcal{R}_{\text{risk}}$, the two terms are not redundant: $\mathcal{R}_{\text{risk}}$ defines the target criterion, while F provides an additional shaping signal that improves credit assignment and trajectory discrimination. Setting $\mu = 0$

recovers a pure risk-sensitive return; setting $\eta = 0$ yields a purely surrogate-ranked objective. GRFR allows both to be combined in a controlled way.

This identifies the total epistemic quality of a trajectory. The first summation is a standard risk-sensitive return: it accumulates task utility while penalizing both intervention cost and latent epistemic risk at each timestep. The second summation introduces an auxiliary shaping term that rewards or penalizes epistemic behavior using the tractable surrogate $F(h_t, y_t)$.

This two-term structure mirrors standard multi-objective and risk-sensitive reinforcement learning, where unobservable or delayed criteria are incorporated through auxiliary cost functionals and shaping terms (Bertsekas, 1995; Tamar et al., 2015). In GRFR, it ensures that trajectories are ranked not only by immediate task performance, but by their cumulative epistemic consequences across the dialogue.

Group-Relative Advantage. To induce comparative learning within each trajectory group, we define a group-relative advantage by subtracting the group mean score:

$$(24) \quad A_i = S(\tau_i) - \frac{1}{K} \sum_{j=1}^K S(\tau_j)$$

This construction is directly analogous to advantage baselines in policy gradient methods, where subtracting a baseline reduces variance and induces learning from relative performance rather than absolute return (Sutton et al., 2000; Schulman et al., 2017). In GRFR, the baseline is not a learned value function but the empirical mean of the trajectory cohort, yielding a group-relative notion of advantage.

In the context of frictive policy discovery, this defines the relative advantage of trajectory τ_i within its group. Rather than using absolute trajectory scores, GRFR evaluates each trajectory by how much better or worse it is than the group average. This implements a purely comparative notion of epistemic quality and removes sensitivity to global score offsets.

By centering scores within each context, GRFR ensures that learning is driven by *which epistemic strategies are better than their alternatives*, rather than by the absolute scale of the trajectory score.

Loss. We update the policy using a normalized, group-weighted policy gradient objective:

$$(25) \quad \mathcal{L}_{\text{GRFR}} = -\mathbb{E} \left[\sum_{i=1}^K \frac{A_i}{\sum_j |A_j|} \log \pi_{\theta}(\tau_i) \right]$$